Unit 5: Quantitative Research Design and Methodology

Lesson 3

Research Instruments for Quantitative Research

Learning Objectives

At the end of the lesson, you should be able to do the following:

- Identify the instruments that researchers use in a quantitative study.
- Create a valid and reliable instrument for a quantitative study.
- Assess the instrument's validity and reliability to provide a good quantitative data.

Read the situation given below. Then, indicate your response on the space provided before the item. Refer to the following responses for each item:

SA = Strongly Agree
A = Agree
N = Neutral
D = Disagree
SD = Strongly Disagree

_____1. I have a large group of friends.

_____2. I enjoy being the center of attention.

_____3. Talking to a stranger never bothers me.

_____4. I like to attend large social events.

_____5. I often get bored whenever I just stay at home.

Types of Research Instruments

An **instrument** can be defined as a tool such as a questionnaire or a survey that measures specific items to gather quantitative data.

Demographic Forms

Demographic forms are used to **collect basic information** such as age, gender, ethnicity, and annual income.

Example of a Demographic Form

1. Age: ___
2. Gender:
   ____ Male
   ____ Female
   ____ Prefer not to say
1. Civil Status:
   ____ Single
   ____ Married
   ____ Widowed
1. Nationality: _____

Performance Measures

Performance measures are used to assess or rate an individual's **ability** such as achievement, intelligence, aptitude, or interests.

Performance Measures

Scenario
In a study investigating the impact of social media usage on students' academic performance, the performance measure would be high if the research assistant consistently and accurately collected data from participants without introducing bias or errors. For example, the research assistant must ensure that data is collected in the same conditions for all participants, preventing external factors like differences in the timing of the surveys, participant fatigue, or distractions from affecting the results.

If the research assistant can address unexpected challenges—such as incomplete surveys or participants misunderstanding the questions—by providing clear instructions and following up for clarification, the performance measure would reflect a strong ability in problem-solving and data integrity management. Additionally, the assistant's skill in identifying and minimizing these issues early ensures the data is reliable and valid for analysis.

Attitudinal Measures

Attitudinal measures are instruments used to measure an individual's **attitudes and opinions** about a subject.

Attitudinal Measures

In a survey assessing students' attitudes toward collaborative learning, the performance measure would be high if the students consistently express positive feelings about working in groups, such as enjoyment of sharing ideas and valuing peer feedback. For instance, if a significant number of students strongly agree with statements like "I feel motivated when I collaborate with my classmates" and "Group discussions enhance my understanding of the material," it indicates a favorable attitude toward collaborative learning. Conversely, if many students express disagreement with such statements, it could signal a negative perception, highlighting the need for interventions to enhance their collaborative experiences.

Behavioral Observation Checklist

Behavioral observation checklists are used to record individuals' **behaviors** and are mostly used when researchers want to measure an individual's actual behavior.

Behavioral Observation Checklist

In a classroom setting where teachers are implementing a new active learning strategy, a behavioral observation checklist can be used to assess how effectively the strategy is being executed. The performance measure would be high if the observers note that a majority of students actively participate in discussions, ask questions, and engage with their peers during activities. For instance, if the checklist includes items like “Students are asking questions,” “Students are working collaboratively,” and “Students are staying on task,” and the observer checks these items frequently, it indicates that the active learning strategy is successfully fostering student engagement and participation. If the behaviors are rarely observed, it may suggest a need for further training or adjustments in the teaching approach.

Factual Information Documents

Factual information documents are accessed to tell information about the participants' **documents**, such as available public records.

Factual Information Documents

In a research project evaluating the effectiveness of a new health intervention, the performance measure would be high if the factual information documents, such as participant consent forms and intervention protocols, are thorough and clearly outline all necessary details. For example, if the documents include comprehensive information on participant eligibility, potential risks, benefits, and the specific procedures involved in the intervention, it demonstrates a commitment to ethical standards and informed consent. Conversely, if the documents lack clarity or essential information, it could hinder participant understanding and compliance, reflecting poor quality in documentation practices that could compromise the integrity of the research.

Which type of research instrument do you think is appropriate for your study? Why?

Constructing Research Instruments for Quantitative Research

How to Construct Research Instruments

1. State your research objectives.
2. Ask questions about your objectives.
3. Gather the required information.
4. Formulate questions.

Tip

The general rule in constructing a research instrument is that it must ensure that the questions in the instrument are relevant to the objectives of the study.

Objectives	Related Questions	Information Required	Questions
1. Explore employees' definition of health and wellness. 2. Identify the strategies to implement the health and wellness program in the company. 3. Develop an objective measure in evaluating an employee's health and wellness.	How do different employees define health and wellness?	Expected activities in promoting health and wellness	1. a. What is your age? 1. b. Are you a ___ male or a ___ female? 1. c. From what department are you? ___ Sales ___ IT ___ Human Resources ___ Executive Office ___ Support
		Perception of health and wellness	2.a. What comes to mind when you hear the words health and wellness?
	In an employee's perspective, what activities promote health and wellness?	Expected activities in promoting health and wellness	3.a. Tick the following activities or programs that you think can promote health and wellness: ___ Yoga exercise ___ Diet Program ___ Others (please specify)

Quality of an Instrument

Reliability

the **stability** and **consistency** of an instrument under different circumstances or points in time.

Validity

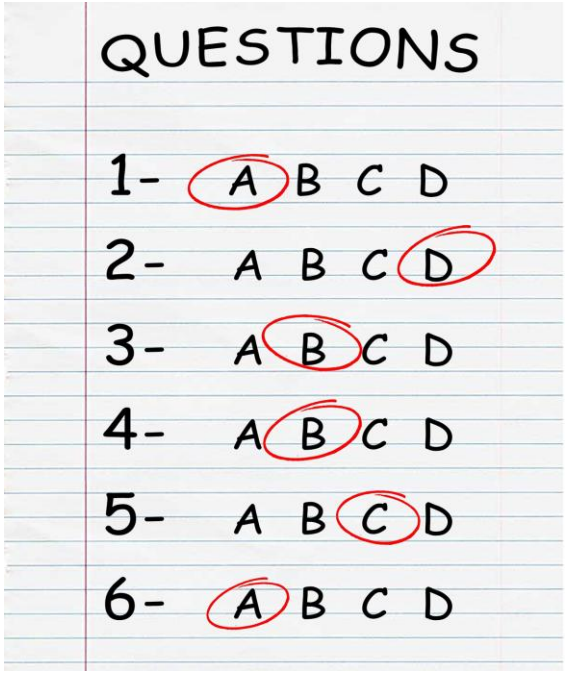
capacity to **measure what it is supposed to measure**.

Do you think a research instrument can be reliable without being valid? Why or why not?

Types of Reliability

1. Internal Consistency

- split-half reliability
- odd-even reliability
- Cronbach's Coefficient Alpha



Types of Reliability

Internal Consistency

In a questionnaire designed to measure student satisfaction with online learning, internal consistency would be assessed by evaluating the extent to which different items that measure the same construct produce similar results. For instance, if the questionnaire includes multiple items like “I feel engaged in my online classes” and “The online format keeps my attention,” a high internal consistency indicates that students who respond positively to one item also tend to respond positively to the others. This suggests that the items are reliably measuring the same underlying construct of student satisfaction.

Types of Reliability

Split Half Reliability

In an educational assessment tool measuring mathematical problem-solving skills, split-half reliability can be evaluated by dividing the test into two halves (e.g., the first half and the second half) and then comparing the scores. For example, if students take a 20-question math test, the first 10 questions may represent the first half, and the last 10 questions represent the second half. If the scores from both halves are highly correlated (e.g., students scoring well on the first half also score well on the second half), it indicates that the test has good split-half reliability, meaning that both halves effectively measure the same skill.

Types of Reliability

Odd Even Reliability

In a personality assessment measuring traits such as extraversion and introversion, odd-even reliability can be checked by comparing scores from odd-numbered questions with those from even-numbered questions. For instance, if the questionnaire has 20 items, the odd-numbered items (1, 3, 5, etc.) and the even-numbered items (2, 4, 6, etc.) should produce similar results if they measure the same trait consistently. A high correlation between the scores from the odd and even items indicates strong odd-even reliability, suggesting that the assessment consistently measures the personality traits.

Types of Reliability

Cronbach's Coefficient Alpha Reliability

In a study measuring self-efficacy in learning environments, researchers can calculate Cronbach's Coefficient Alpha to determine the reliability of the self-efficacy scale. For example, if the scale consists of 10 items that assess various aspects of self-efficacy, such as "I can complete my assignments on time" and "I am confident in my ability to understand new topics," a Cronbach's Alpha of 0.85 would indicate excellent internal consistency. This means that the items in the scale are reliably measuring the same construct of self-efficacy, and the scale is likely to yield consistent results across different populations.

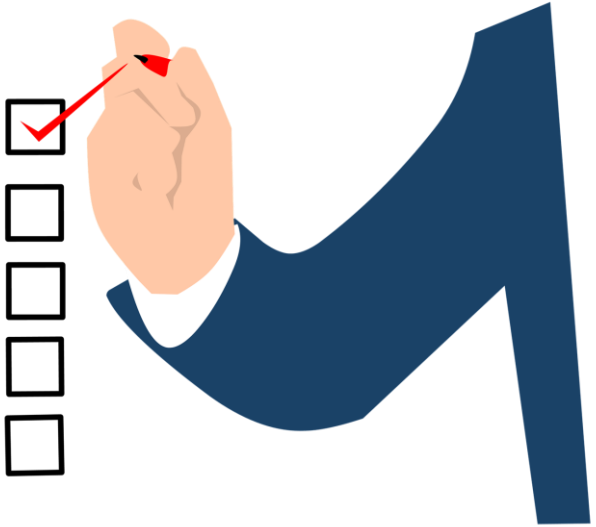
Did you know?

The **Cronbach's coefficient alpha** may be obtained using computer softwares like statistical analysis softwares or spreadsheets.

Types of Reliability

2. **Stability Over Time**
   test-retest reliability

3. **Alternate Forms**
   also called as parallel forms



Types of Reliability

Test – Retest Reliability

In a psychological study measuring anxiety levels, researchers can assess test-retest reliability by administering the same anxiety questionnaire to participants at two different points in time (e.g., one week apart). For instance, if a participant scores 25 on the anxiety scale during the first test and again scores 24 on the second test, this consistency in scores suggests high test-retest reliability. It indicates that the anxiety measurement is stable over time, meaning that an individual's anxiety level has not significantly changed within that week and that the instrument reliably captures the trait.

Types of Reliability

Alternate Forms (Parallel Forms)

In an educational setting, a school district may use an alternate forms reliability approach to evaluate students' mathematics skills. For example, two equivalent versions of a math test (Form A and Form B) are created, both assessing the same skills but with different questions. Students take Form A at the beginning of the term and then take Form B later in the term. If the scores from both forms are highly correlated (e.g., students who perform well on Form A also perform well on Form B), it indicates that the two forms are measuring the same construct consistently. This demonstrates good alternate forms reliability, ensuring that different versions of the test yield similar results.

What particular constructs or variables are most likely to be stable over time?

Types of Validity

Face Validity

extent to which an instrument appears to measure what it is supposed to measure

Content Validity

Content validity refers to how well a research instrument covers all the aspects of the concept being studied.

Types of Validity

Face Validity

- Example: A survey on **anxiety** contains questions about nervousness, worry, and physical symptoms like sweating or trembling. If people who take the survey feel that the questions directly relate to anxiety, the survey has good face validity. While face validity is not scientifically rigorous, it is often important for user acceptance.

Types of Validity

Content Validity

- **Example:** A researcher designs a survey to measure **job satisfaction** among employees. To ensure **content validity**, the researcher makes sure that the survey covers all dimensions of job satisfaction, such as pay, work environment, relationships with colleagues, career growth, and job security. The researcher may consult experts or use literature reviews to ensure that the survey questions address every relevant aspect of job satisfaction.

Types of Validity

Extenal Validity

External validity refers to the extent to which the results of a study can be generalized to other settings, populations, or times.

Internal Validity

Internal validity refers to the extent to which a study can demonstrate that changes in the dependent variable are due to the independent variable, rather than confounding factors or alternative explanations.

Types of Validity

External Validity

- **Example:** A researcher conducts a study on **college students' study habits** at one university. To establish **external validity**, the researcher would need to show that the findings apply not just to students at this particular university, but also to college students in other universities, or even students in other educational levels. The study would have higher external validity if the results could be generalized across different contexts.

Types of Validity

Internal Validity

- **Example:** In an experiment testing whether **a new teaching method improves student performance**, **internal validity** would be high if the researcher can confidently say that the improved performance was due to the teaching method, and not other factors like students' prior knowledge, differences in teacher ability, or external distractions during the experiment.

Types of Validity

Criterion Validity

Criterion validity refers to how well a measurement correlates with a relevant external criterion, which can be divided into two types: **concurrent** and **predictive** validity.

Concurrent Validity

This type of validity examines whether the measurement correlates with another measure taken at the same time.

Types of Validity

Predictive Validity

This refers to how well a measurement predicts future outcomes or behaviors.

Types of Validity

Concurrent Validity

- **Example:** A researcher administers a new test for **mathematical ability** to high school students. To establish **concurrent validity**, the researcher compares the results of this new test with the students' grades in their current math classes. If there is a strong correlation between the test scores and their grades, the test has strong concurrent validity.

Types of Validity

Predictive Validity

- **Example:** A company uses a new **aptitude test** to hire employees. To assess **predictive validity**, the researcher tracks the performance of the hired employees over the next year. If the scores on the aptitude test are good predictors of job performance, the test has high predictive validity.

Remember

A valid test is always reliable but a reliable test is not always valid.

Quick Check

Bianca is a preschool teacher. She doesn’t use numerical grades for her students given that the level she is teaching involves identifying developmental milestones. Instead, she prepares a list of actions that her students were already able to accomplish.

Answer:

Behavioral observation checklist

Quick Check

Renzo is a restaurant manager, who recently opened another branch in Quezon City. He needs to hire new personnel for his restaurant. He designed an instrument, which measures work indicators of prospective employees' likelihood to succeed in their job.

Answer:

Performance measures

Quick Check

Why is it important to establish first the objectives of a study to determine the quality of the research instrument to be used?

The objectives or the research questions that the study aims to answer play a big factor in deciding which research instrument is appropriate for the study because it gives the researcher an idea about what specific variables or construct must be explored. This also determines the types of questions that should be asked to the participants in order to fulfill the objectives of the research study.

Wrap Up

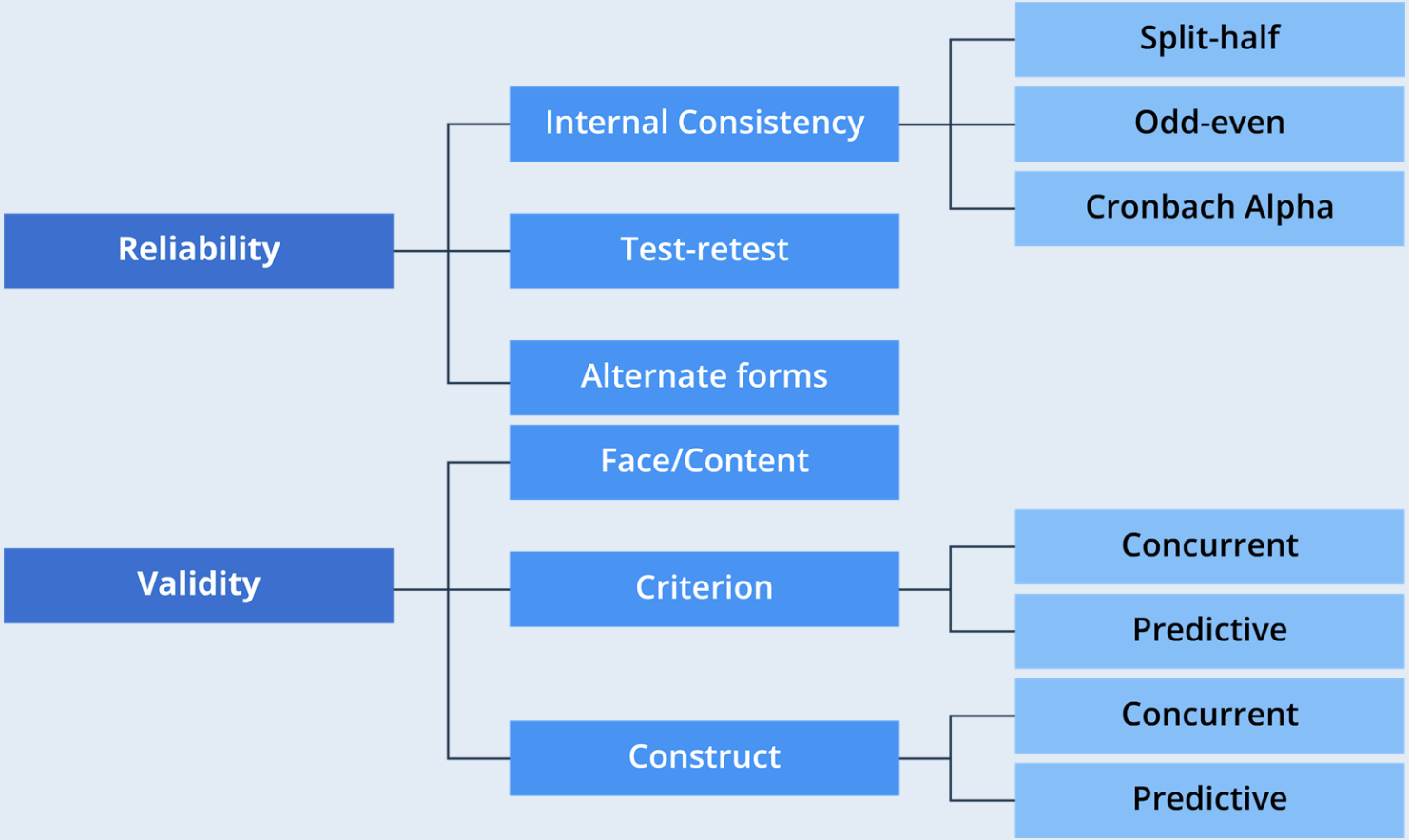
- Research instruments for quantitative studies mostly have **five general types**: demographic forms, performance measures, attitudinal measures, behavioral observation checklists, and factual information documents.

- The quality of a research instrument can be determined by its **reliability** and **validity**.

- **Reliability** is defined as the stability and consistency of an instrument over a period of time.

- **Validity** pertains to the extent where an instrument accurately measures what it is supposed to measure.

Wrap Up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